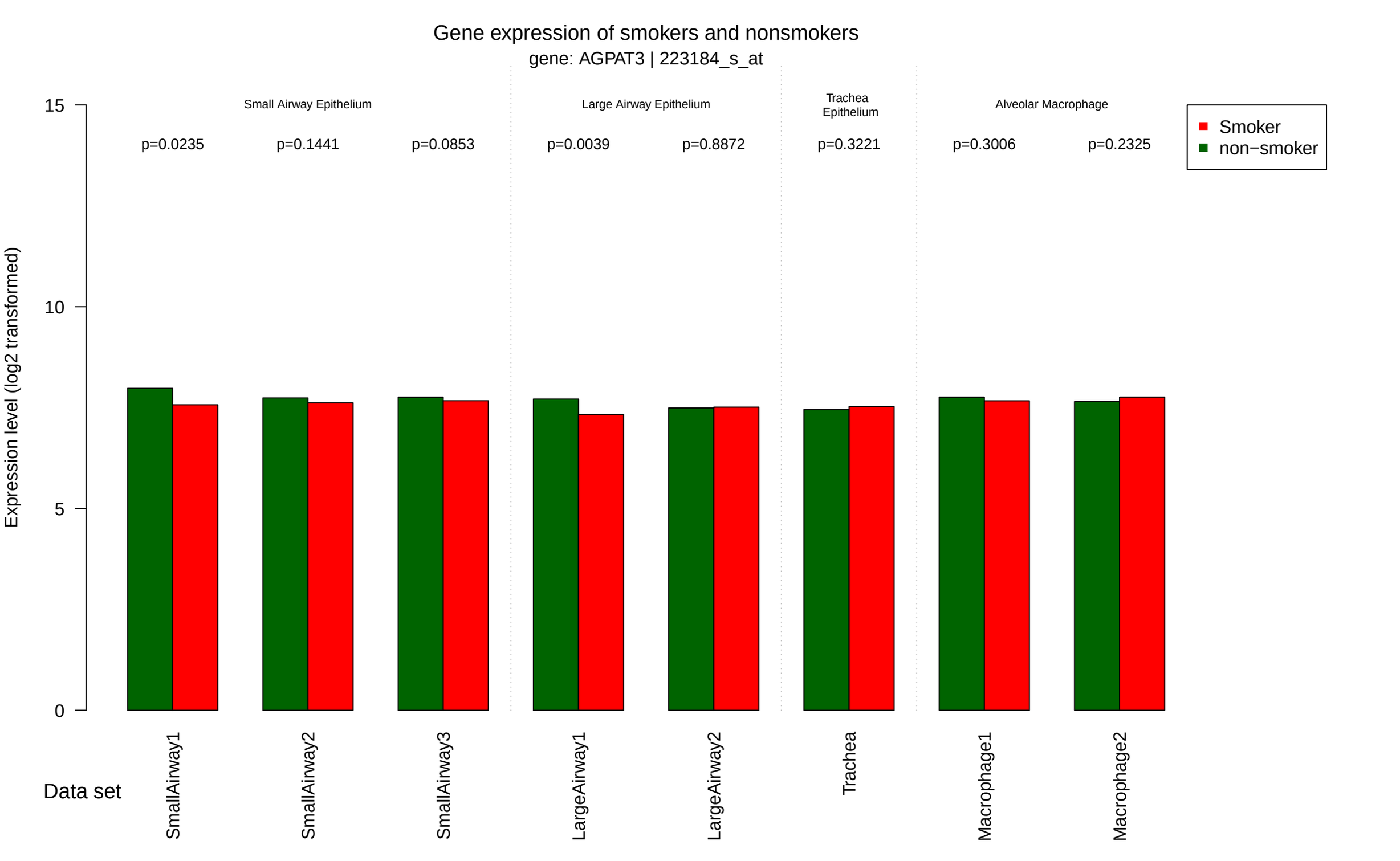
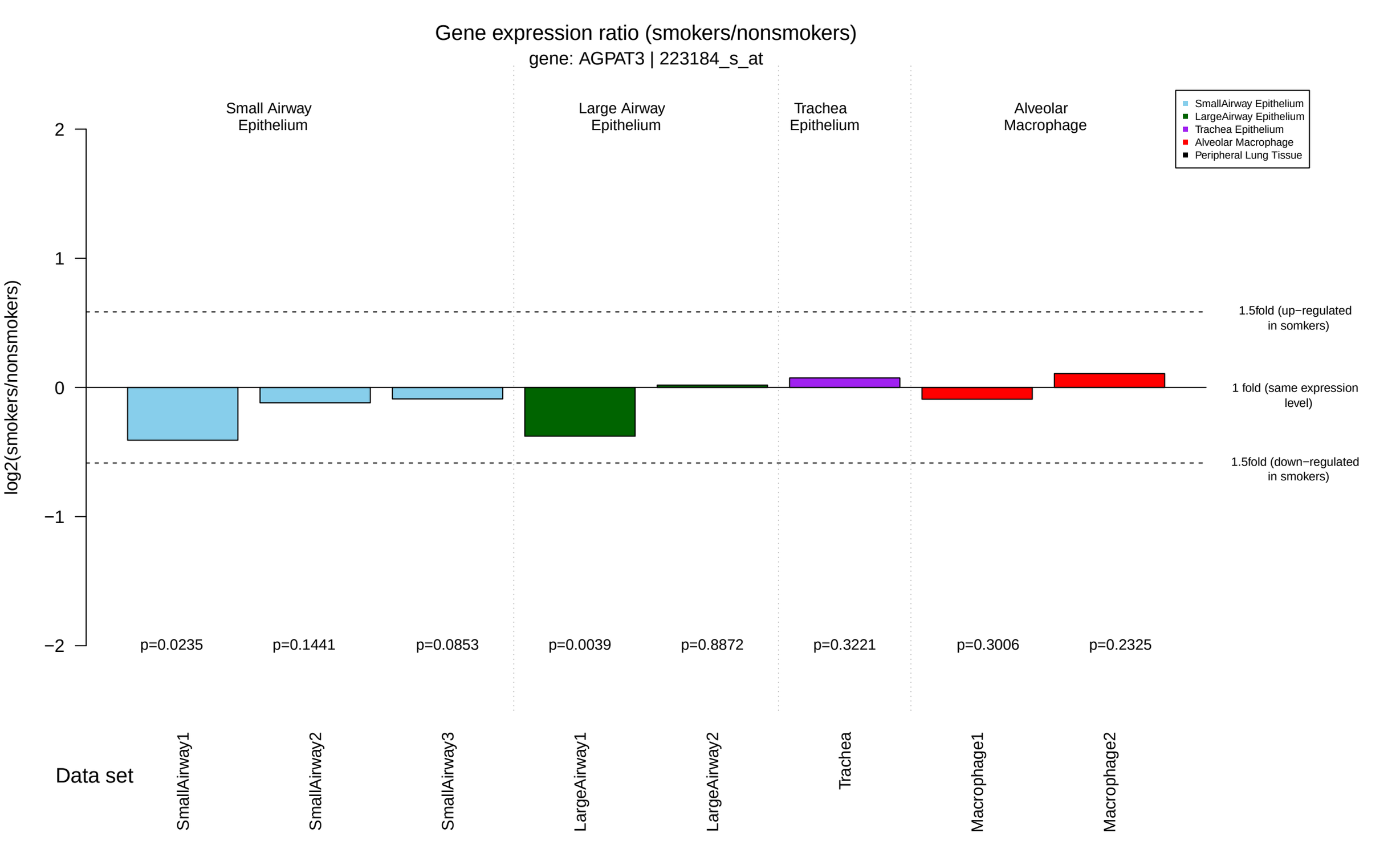
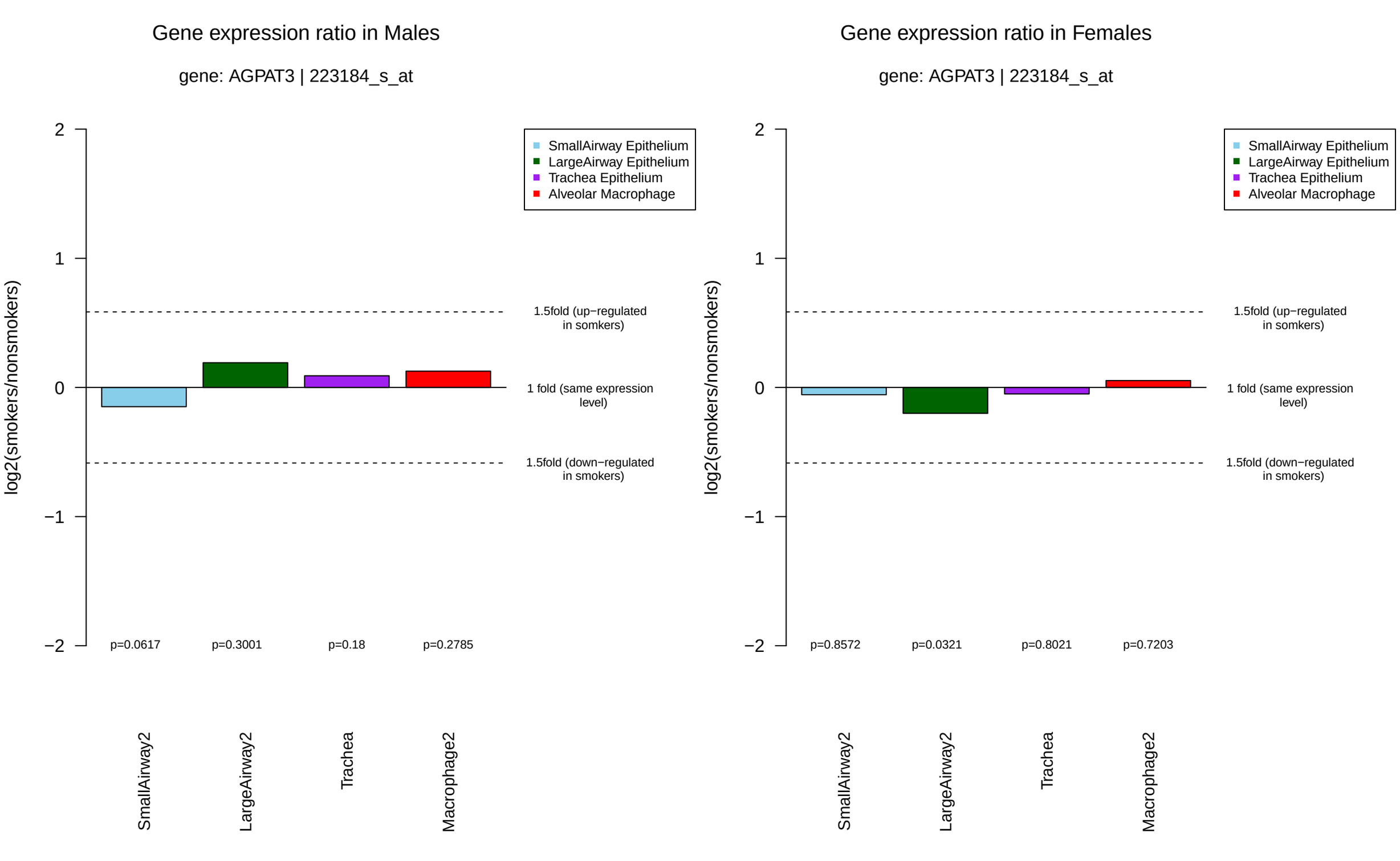
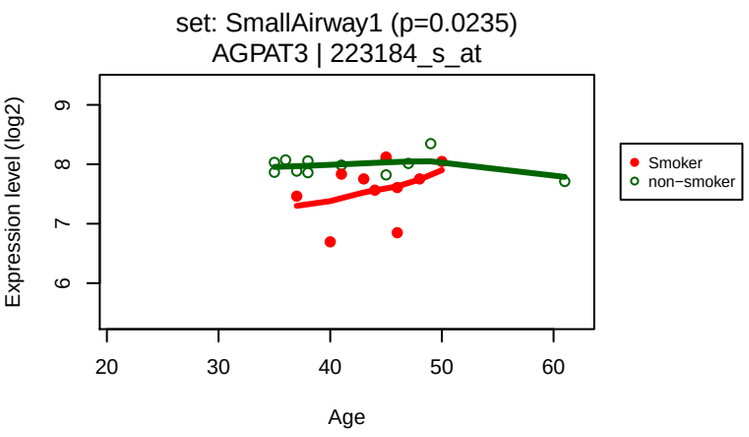
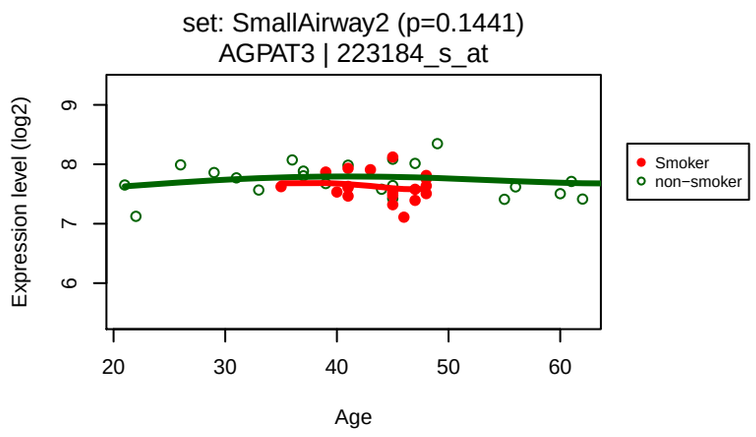
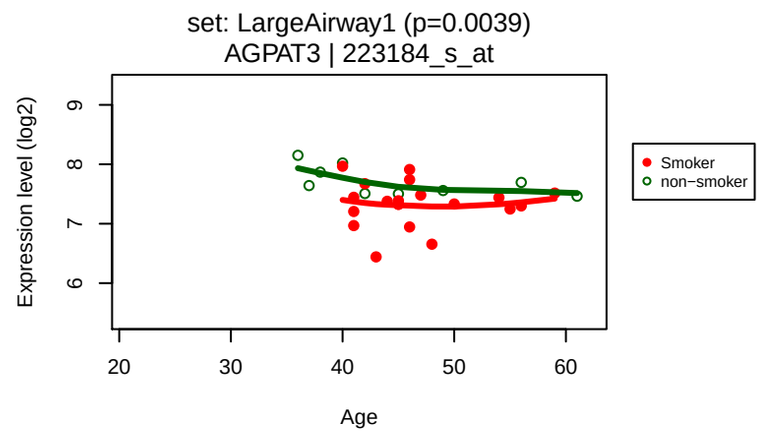

Gene expression of smokers and nonsmokers
gene: AGPAT3 | 223184_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0235
p=0.1441
p=0.0853
p=0.0039
p=0.8872
p=0.3221
p=0.3006
p=0.2325
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

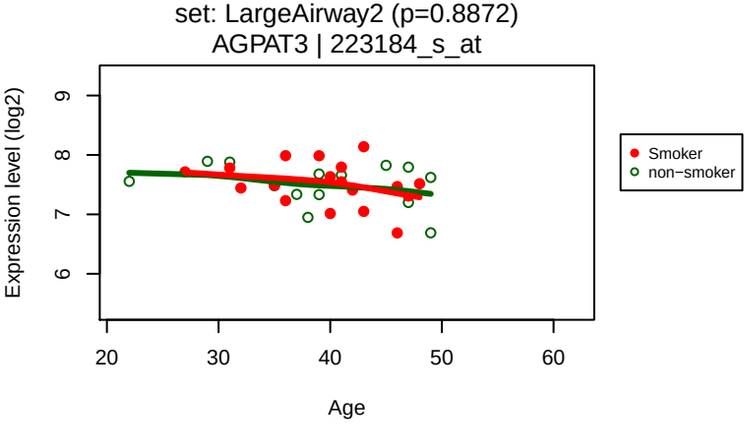

Gene expression ratio (smokers/nonsmokers)
gene: AGPAT3 | 223184_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0235
p=0.1441
p=0.0853
p=0.0039
p=0.8872
p=0.3221
p=0.3006
p=0.2325
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

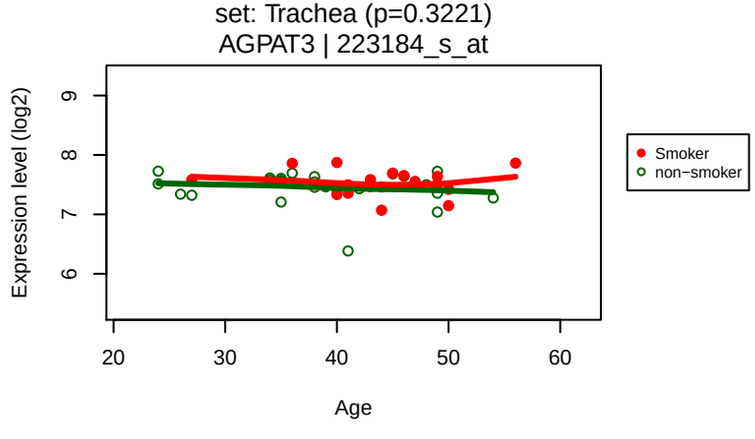

Gene expression ratio in Males
gene: AGPAT3 | 223184_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0617
p=0.3001
p=0.18
p=0.2785
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AGPAT3 | 223184_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8572
p=0.0321
p=0.8021
p=0.7203
SmallAirway2
LargeAirway2
Trachea
Macrophage2

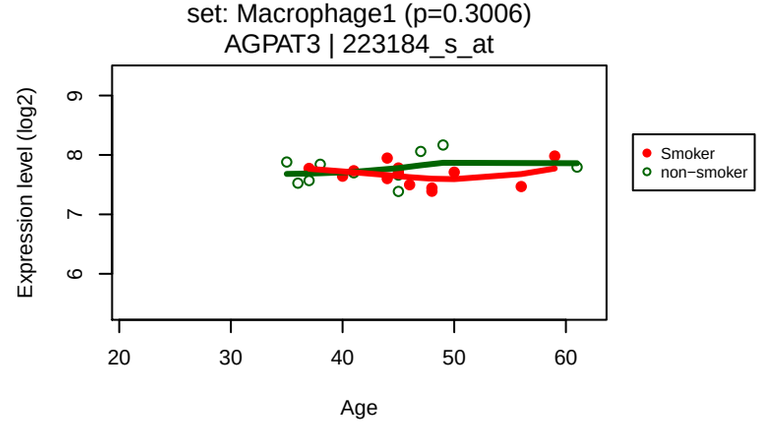

set: SmallAirway1 (p=0.0235)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker

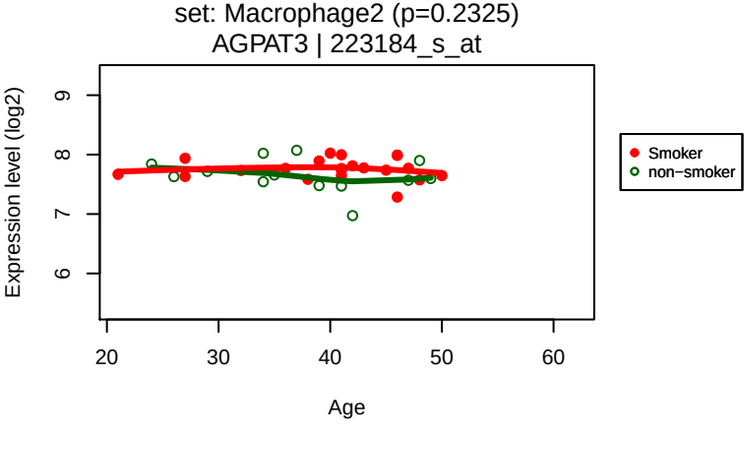

set: SmallAirway2 (p=0.1441)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker

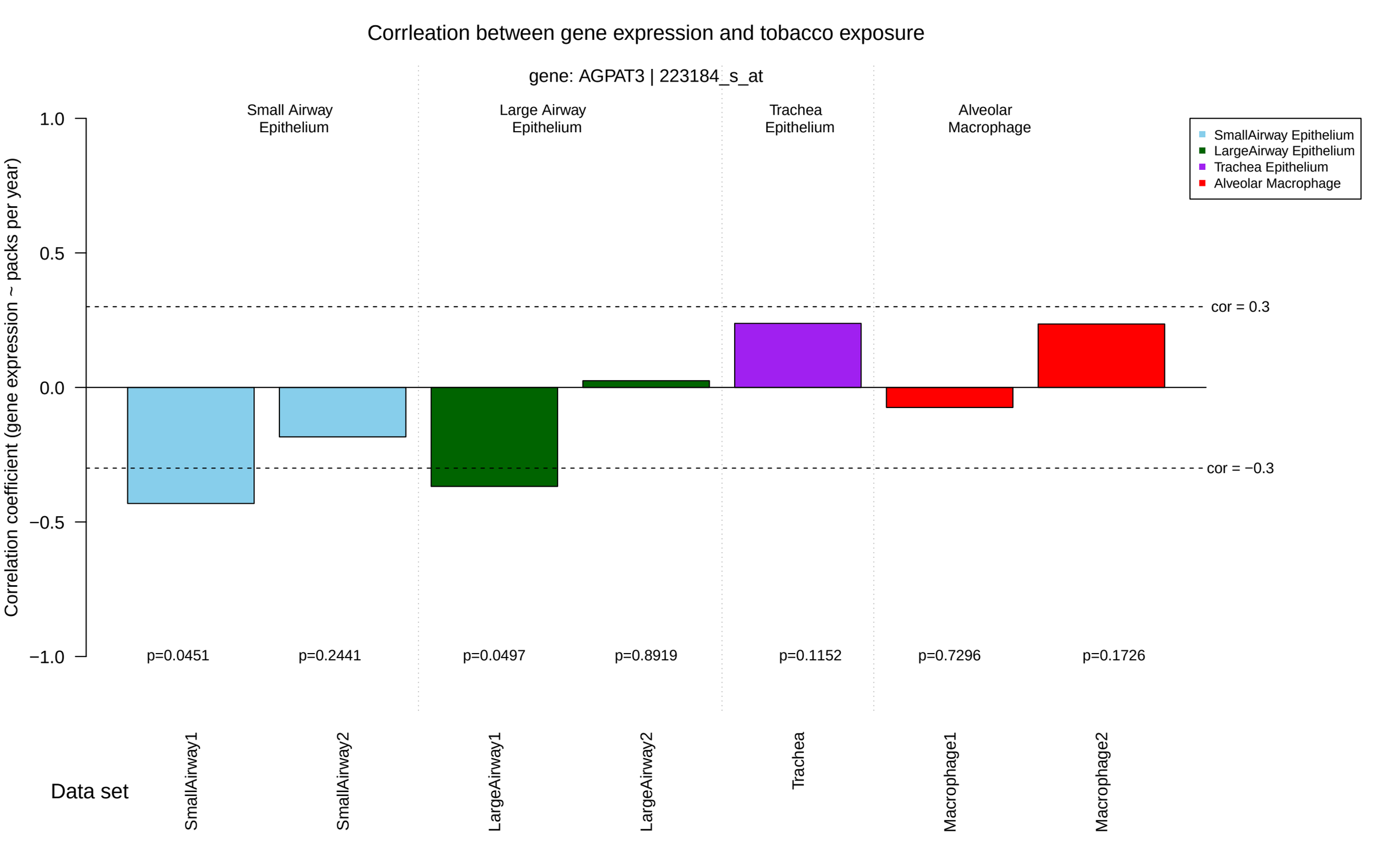

set: LargeAirway1 (p=0.0039)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8872)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3221)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3006)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2325)
AGPAT3 | 223184_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AGPAT3 | 223184_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0451
p=0.2441
p=0.0497
p=0.8919
p=0.1152
p=0.7296
p=0.1726
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2